Hackathon Challenge Pack

1. Digital Fraud Investigation System

Level 1: Detect, Analyse and Flag Suspicious Digital Activity

Suggested time: 0 -3 hours

Objective: Build a working fraud detection application that ingests digital financial activity, identifies suspicious patterns and presents explainable fraud alerts for investigation.

Required Capabilities

1. Data Ingestion and Integration

Build an application that allows users to upload or select data from different digital financial systems. The application should support the following sample data:

- Customer and account profiles containing customer identifiers, account details, contact information and onboarding information.
- Transaction batches containing amounts, timestamps, sender and beneficiary details, transaction channels and statuses.
- Device and login logs containing device identifiers, IP addresses, login timestamps, authentication outcomes and device changes.

The application should accept common structured formats such as CSV, Excel and JSON. It must validate uploaded records, identify missing mandatory information and avoid silently ignoring invalid data. Participants should use a common data structure that allows records from different systems to be linked through customer, account, application, device or transaction identifiers.

2. Fraud Detection Rules and Behavioural Analysis

Implement a detection mechanism that identifies suspicious activities using configurable rules, statistical analysis, machine learning or a combination of approaches. At a minimum, the application must detect the following four scenarios:

Scenario	Expected detection behaviour
Suspicious transfers	Identify unusual amounts, rapid transfers, new beneficiaries or transactions inconsistent with customer history
Shared devices	Identify multiple customer profiles using the same device and assess whether the relationship is suspicious
Repeated login failures	Detect repeated unsuccessful logins, particularly when followed by a successful login or sensitive account activity
Legitimate lookalikes	Avoid incorrectly classifying ordinary customer behaviour as fraud merely because it resembles a suspicious pattern

Participants should also implement at least one additional fraud scenario from the challenge catalogue, such as suspicious loan applications or duplicate insurance claims.

3. Fraud Risk Assessment

Every detected suspicious activity should receive a risk classification.

The application may use a risk score, such as 0–100, or clearly defined categories such as Low, Medium and High. The score should consider multiple relevant indicators rather than relying only on a single threshold.

For example, a high-value transfer should not automatically be considered fraudulent if the customer regularly performs similar transactions.

However, a high-value transfer immediately following multiple failed login attempts and a new-device login may warrant greater scrutiny. The application must explain the factors contributing to the assigned risk level.

4. Fraud Alert Dashboard

Develop a dashboard where users can inspect detected suspicious activities.

The dashboard should display the number of processed records, flagged events, risk distribution and fraud categories. Users must be able to filter alerts by customer, transaction, product, risk level, date and detection type.

Selecting an alert should display the underlying event, the reasons it was flagged and the supporting records.

5. Basic Fraud Case Queue

Each actionable fraud alert should be available in a case queue.

A case should contain a unique identifier, affected customer or account, suspected fraud category, risk level, supporting evidence and investigation status. Users should be able to open the case and record an initial decision, such as Requires Investigation or False Positive. The system must preserve the original alert even when its classification is subsequently changed.

Expected Submission

A working application demonstrating data ingestion, fraud detection, risk assessment, an alert dashboard and a basic investigation queue.

Level 1 must be submitted and accepted before Level 2 can begin.

Level 2: Investigate, Correlate and Manage Fraud Cases

Suggested time: 3 - 5.5 hours

Objective: Extend the detection application into an intelligent fraud investigation system that connects related activities across financial systems, supports analyst decisions and recommends appropriate actions.

Required Capabilities

1. Cross-System Fraud Correlation

Build a mechanism that links suspicious activities across lending, insurance and core banking systems.

For example, a customer may submit a loan application from a device associated with multiple identities, receive loan proceeds into a recently changed bank account and transfer the funds to an unfamiliar beneficiary.

The system should connect these activities and present them as a related investigation rather than three unrelated alerts. It should support relationships involving customers, accounts, devices, IP addresses, phone numbers, applications, transactions, policies and claims.

2. Fraud Relationship Analysis

Introduce a visual representation of related fraud events and entities. For example:

- Illustrative investigation network showing relationships between customers, devices, applications and financial transactions.
- The system should allow investigators to select a connected entity and inspect its associated records.
- Relationships must be supported by actual matching identifiers or documented evidence, not invented by the AI model.

3. Intelligent Fraud Case Consolidation

Multiple alerts relating to the same suspected fraud activity should be grouped into a single investigation where appropriate.

For example, repeated failed logins, a successful login from a new device and a suspicious transfer may be consolidated into one account takeover investigation. The system should explain why alerts were grouped and allow an analyst to separate them if the relationship is incorrect. It must retain the original alerts and their evidence.

4. Fraud Investigation Workspace

Develop a workspace where an investigator can examine a case in detail. The workspace should include a timeline of events, affected entities, fraud indicators, supporting evidence, risk assessment and investigation notes. The analyst should be able to:

- Confirm that the activity requires further investigation.
- Mark the alert as a false positive with a reason.
- Request additional information.
- Escalate the case to a fraud specialist or supervisor.
- Recommend a temporary hold or enhanced verification.
- Record an investigation conclusion.

Actions that affect customer accounts or transactions must remain recommendations unless explicitly approved through an authorised workflow.

5. Evidence Management and Audit Trail

Every case must maintain an evidence history. This should include the source records, detected indicators, AI-generated assessments, analyst corrections, decisions, timestamps and user identity. If an analyst changes a risk classification or dismisses an alert, the system must retain the original classification and record why the decision was changed. Evidence should be downloadable as a structured case report.

6. Fraud Prioritisation

Implement a mechanism to prioritise cases according to risk and potential financial impact. For example, a suspicious transfer involving a large amount and recent account takeover indicators may be prioritised over an isolated failed login. The application should consider severity, financial exposure, number of related events and confidence in the detection. Cases with insufficient evidence should be marked for further review rather than automatically assigned the highest risk.

Expected Submission

An integrated fraud investigation application with cross-system correlation, relationship visualisation, consolidated cases, analyst workflows and evidence management.

Level 2 must be submitted and accepted before Level 3 can begin.